

Wang Zixuan

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EDUCATIONAL BACKGROUND

Hong Kong University of Science and Technology (Guangzhou), AI Thrust

09/2025 – Present

MPhil in Artificial Intelligence

GPA 4.00

- Working with Prof. Qin Chengwei on large language models, agent memory and multi-agent systems.
- Relevant coursework: Advanced Machine Learning, Reinforcement Learning, Natural Language Processing and Distributed Systems.

Hong Kong Baptist University

09/2021 – 06/2025

BSc(Hons) in Data Science

GPA 3.59, First Prize Scholarship

- Selected courses: Data Mining, Design and Analysis of Algorithms, Natural Language Processing, Calculus for Science and Engineering, Linear Algebra, Discrete Structures, Data Structures and Algorithms and Computer and Network Security.

PREPRINTS & UNDER REVIEW

- **BizCompass: Benchmarking the Reasoning Capabilities of LLMs in Business Knowledge and Applications** (ACL 2026 conference submission). Constructed a comprehensive benchmark to evaluate the reasoning capabilities of large language models on business knowledge and application tasks; analysed results across models and identified improvement directions.
- **Unified-MAS: Universally Generating Domain-Specific Nodes for Empowering Automatic Multi-Agent Systems** (ACL ARR 2026 March submission). Proposed a unified approach for automatically generating domain-specific nodes to empower multi-agent systems across tasks and domains.
- **TiARA: Time-Accumulated Rank Adaptation for Constant-Cost Fine-Tuning** (ICML 2026 conference submission). Developed a time-accumulated rank adaptation technique for constant-cost fine-tuning, reducing computational requirements while maintaining model performance.
- **AMA: Adaptive Memory via Multi-Agent Collaboration** (ACL ARR 2026 January submission). Introduced an adaptive memory framework through multi-agent collaboration to improve memory retrieval and utilisation in large language models.
- **Co-Learning for Multi-Agent Reinforcement Collaborative Framework with Conversational Natural Language Interfaces** (under review at JAAMAS). Proposed a learning framework where multiple agents communicate via natural language and jointly learn policies.
- **Unnoticeable Graph Injection Attack on Graph Neural Networks** (under review at DSN). Investigated stealthy graph injection attacks on GNNs; developed *UNGIA*, a method that selects representative nodes and employs bi-level optimisation to balance homogeneity and attack effectiveness.

COMPETITIONS & HONOURS

Kaggle – LLM Prompt Recovery Competition

02/2024 – 04/2024

Bronze Medal

- Participated in a competition focused on recovering prompts used to transform given text, employing a pre-trained causal language model (Mistral 7B v2) to generate cue words for rewriting tasks.
- Quantified and optimised token handling, discovering that special tokens pull sentences into specific focal points in the embedding space and highlight key information.

Mathematical Contest in Modeling (MCM)

02/2023 – 05/2023

Honorable Mention

- Analysed and established a mathematical model for the *New York Times* “fill-in-the-blank” game data to predict the game’s development trends.
- Predicted the rise and fall of the game based on past participant numbers, question difficulty and response times; provided suggestions on adjusting difficulty to increase participation and considerations for word selection.

Mathematical Contest in Modeling (MCM)

02/2022 – 05/2022

Honorable Mention

- Investigated price fluctuations of assets (gold, USD and stocks) over five years; studied various information to predict future developments.
- Formulated an investment plan and calculated returns according to predictions; enhanced understanding of time series analysis, investment strategies and statistical programming.

INTERNSHIPS & WORK EXPERIENCE

Guangdong Province Key Laboratory of Interdisciplinary Research and Application for Data Science

05/06/2024 – 05/09/2024

Research Assistant, Zhuhai, China

- Developed a model architecture supporting incremental learning based on the *Learning to Edit* long-term memory mechanism, addressing knowledge editing issues when medical data are dynamically updated.
- Assisted in building LLM agents for medical scenarios, integrating text, image and structured medical record data; the goal was to achieve an automated triage suggestion system and patient Q&A interaction system, and to optimise reasoning paths in complex decision chains via reinforcement learning.
- Aided the supervisor in model fine-tuning; used LangChain to design workflows for medical agents and combined Spark to implement distributed preprocessing and privacy-protection for multi-source heterogeneous medical data.

Vivo Mobile Communication Co., Ltd

06/06/2023 – 04/09/2023

Data Analyst Manager, Shenzhen, China

- Completed over 20 analyses related to existing products, generated more than 40 insights and recommendations and implemented over 10 initiatives, cumulatively driving an increase of 110,000 daily active users.
- Decomposed data targets for the marketing team’s new product development, establishing measurement methodologies for sub-products; built dashboards accessed over 800 times by colleagues.
- Participated in the framework train for the company’s data assistant LLM model; developed a Python-based demo and collaborated with front-end and back-end teams to finalise the online version.
- Conducted capability assessments in SQL, Python analysis and visualisation coding; contributed to optimising the data assistant framework and improved single-table SQL generation accuracy from around 45% to 62–67%; achieved an 87% matching rate and 89% successful charting rate for visualisation code.

SELECTED RESEARCH PROJECTS

Research on the Stealthiness of Injection Attacks in Graph Neural Networks

06/2024 – 09/2024

Graduation project supervised by Dr. Donglong Chen

- Proposed **UNGIA**, a stealthy graph injection attack method utilising clustering techniques to identify representative nodes.

- Conducted experiments on benchmark datasets (Cora, Citeseer and Cora-ML) to demonstrate the advantages of UNGIA.
- Selected nodes as anchor points to inject fake nodes, trained a generator to learn the distribution of node features and introduced bi-level optimisation to balance homogeneity constraints and attack effectiveness.

Chaos-Based GNN for Aspect-Based Sentiment Analysis

11/2023 – 09/2024

Collaborative project

- Analysed current trends and challenges in natural language processing and aspect-based sentiment analysis on social media.
- Developed and optimised models to handle polysemy, semantic correlation and emotional intricacies; collected large-scale datasets and designed additional experiments to refine algorithms.
- Collaborated with researchers from computer science, NLP and psychology to advance complex sentiment analysis techniques.

CNO-former: Chaotic Neural Oscillation Self-Attention Model

09/2023 – 09/2024

Research project

- Formulated **CNO-former**, a neural network model based on the Transformer that modifies neuron states within the self-attention mechanism to reduce computational cost and mitigate information loss.
- Introduced Lee-oscillators to enhance chaotic multi-head and masked multi-head self-attention in both encoder and decoder.
- Demonstrated that CNO-former integrates chaos theory and self-attention mechanisms to optimise efficiency, accelerate convergence and enhance representation capabilities; awarded an Excellent Award in the 2nd Computer Science Related Poster Exhibition (one of only three projects awarded).

Advanced Multi-Agent Systems Development

05/2023 – 02/2025

Guangdong Provincial Special Scientific Research Program

- **Multi-Agent Teaching Community System Based on Reinforcement Learning:** formulated a teaching agent system integrating the CNO-former model and embedding a knowledge vector database; employed language model embeddings to transform text into feature vectors with positional information and established a management framework for coordinating multiple agents.
- **Multi-Agent Tour Guide Assistance System:** developed a system for a small island in Guangdong that provides personalised tour guide services; collected and cleaned tourism resource data to optimise user experiences.